

# Melanie Subbiah

Somerville, MA

✉ [m.subbiah@columbia.edu](mailto:m.subbiah@columbia.edu)

🌐 <https://melaniesubbiah.github.io>

## Education

- 2020 – present  **Ph.D. Computer Science, Columbia University** (*in progress*)  
Topic: Computational understanding of subtext in writing  
Advisor: Kathleen McKeown
- 2020 – 2024  **M.Phil. Computer Science, Columbia University**  
Completed as part of PhD.
- 2020 – 2022  **M.S. Computer Science, Columbia University**  
Completed as part of PhD.
- 2013 – 2017  **B.A. Computer Science, Williams College**, Magna Cum Laude, Phi Beta Kappa  
Thesis: *Using Text Abstraction and LSTM Language Models for Domain-Independent Narrative Generation*  
Advisor: Andrea Danyluk

## Work Experience

- 2021-2023  **Columbia University** New York, NY  
*Graduate Research Assistant*  
Research in Kathleen McKeown's lab as part of my PhD.
- (Jun.–Aug.) 2022  **Meta** New York, NY  
*Machine Learning Research Intern - AI for Augmented Reality Input & Interaction*  
Deep learning to interpret EMG data for human-computer neural interfaces.
- 2019 – 2020  **OpenAI** San Francisco, CA  
*Member of Technical Staff - Language*  
Evaluation suite for GPT-3, co-first-author on the GPT-3 paper (Language Models are Few-Shot Learners).
- 2017 – 2019  **Apple** Cupertino, CA  
*Machine Learning Engineer - AI Research*  
Data center HVAC efficiency, reward function design for autonomous systems, domain randomization for sim-to-real transfer in computer vision, and effective QA methods for crowdsourced annotated data.
- (Jun.–Aug.) 2016  **Facebook** Menlo Park, CA  
*Software Engineer Intern - Site Efficiency*  
Internal C++ tool to monitor/visualize the efficiency of multi-threaded functions.
- (Jun.–Aug.) 2015  **Fathom Information Design** Boston, MA  
*Data Science Intern - Activity Characterization*  
Clustering and visualizing activity patterns in daily movement data from wearables.
- (Jun.–Aug.) 2014  **Dartmouth College** Hanover, NH  
*Research Assistant - Epidemiology*  
Statistical analysis for study of in utero arsenic exposure.

## Honors and Awards

---

- 2025
  - **Best Paper** (for papers published in 2024), *TACL*  
Awarded for my first-author paper, "Reading Subtext: Evaluating Large Language Models on Short Story Summarization with Writers."
- 2024
  - **Outstanding Reviewer**, *EMNLP*  
Recognized as an outstanding reviewer for my thoughtful reviews and productive discussion with authors.
  - **Most Influential Scholar Award Honorable Mention**, *AMiner AI 2000*  
Recognized as a top 100 most impactful machine learning scholar between 2014 and 2023.
- 2023
  - **Top 100 AI Achievements from 1943-2021**, *BenchCouncil AI100*  
Recognized for my work on the GPT series of models.
  - **PhD Fellowship**, *Amazon/Columbia Center of Artificial Intelligence Technology*  
Funding for two years of the PhD awarded by the Columbia/Amazon AI Center.
- 2021
  - **PhD Fellowship Honorable Mention**, *NSF Graduate Research Fellowship Program*  
Awarded an honorable mention for my submission.
- 2020
  - **Outstanding Paper**, *NeurIPS*  
Awarded for my co-first-author paper, "Language Models are Few-Shot Learners" (GPT-3 paper).
  - **PhD Fellowship**, *Columbia Presidential and SEAS Fellowship*  
Columbia University funding award for incoming PhD students.
- 2017
  - **Phi Beta Kappa Student Speaker**, *Williams College Commencement*  
Voted by the Phi Beta Kappa students to speak at Williams College graduation.
  - **Highest Honors & Best Colloquium Presentation**, *Williams College Computer Science*  
Williams College computer science department awards for the graduating class. Awarded for my work on my senior thesis.
  - **Shorty Story Writing Honorable Mention**, *Williams College Benjamin B. Wainwright Prize*  
Williams College writing competition for short stories.

## Publications

---

### Conference Proceedings/Findings and Journals

- [INLG 25] Limpijankit, M., Chen, Y., [Subbiah, M.](#), Deas, N., & McKeown, K. (2025). Counterfactual Simulatability of LLM Explanations for Generation Tasks. In *Proceedings, International Natural Language Generation*.
- [EMNLP 25] [Subbiah, M.](#), Mishra, A., Grace Kim, L. T., Durrett, G., & McKeown, K. (2025). Is the Top Still Spinning? Evaluating Subjectivity in Narrative Understanding. In *Proceedings, Empirical Methods in Natural Language Processing*.
- [TACL 24] [Subbiah, M.](#), Zhang, S., Chilton, L. B., & McKeown, K. (2024). Reading Subtext: Evaluating Large Language Models on Short Story Summarization with Writers. In *Transactions of the Association for Computational Linguistics*, MIT Press (**Best Paper Award**).
- [EMNLP 24] [Subbiah\\*](#), [M.](#), Ladhak\*, F., Mishra, A., Adams, G., Chilton, L. B., & McKeown, K. (2024). STORYSUM: Evaluating Faithfulness in Story Summarization. In *Proceedings, Empirical Methods in Natural Language Processing*.
- [ACL 23] Storek, A., [Subbiah, M.](#), & McKeown, K. (2023). Unsupervised Selective Rationalization with Noise Injection. In *Proceedings, Association for Computational Linguistics*.
- [ACL 23] Wang, G., Chillrud, L., Harwood, K., Ananthram, A., [Subbiah, M.](#), & McKeown, K. (2023). Check-COVID: Fact-Checking COVID-19 News Claims with Scientific Evidence. In *Findings, Association for Computational Linguistics*.

- [EMNLP 22 ] Levy, S., Allaway, E., **Subbiah, M.**, Chilton, L., Patton, D., McKeown, K., & Wang, W. (2022). SafeText: A Benchmark for Exploring Physical Safety in Language Models. In *Proceedings, Empirical Methods in Natural Language Processing*.
- [EMNLP 22 ] Mei, A., Kabir, A., Levy, S., **Subbiah, M.**, Allaway, E., Judge, J., ... Wang, W. (2022). Mitigating Covertly Unsafe Text within Natural Language Systems. In *Findings, Empirical Methods in Natural Language Processing*.
- [NeurIPS 20 ] Brown\*, T., Mann\*, B., Ryder\*, N., **Subbiah\*, M.**, Kaplan, J. D., Dhariwal, P., ... Amodei, D. (2020). Language Models are Few-Shot Learners. In *Proceedings, Neural Information Processing Systems (Outstanding Paper Award)*.
- [PLOS One 17 ] Nygaard, U., Li, Z., Palys, T., Jackson, B., **Subbiah, M.**, Malipatlolla, M., ... Nadeau, K. (2017). Cord blood T cell subpopulations and associations with maternal cadmium and arsenic exposures, PLoS One.

## Refereed Workshops

- [ARP Symposium 25 ] Mayukha, A., **Subbiah, M.**, Guzman, A., Jitklongsab, S., & McAdams, D. (2025). *Methodological Approaches to the Analysis of Life Story Interviews*. Symposium at the Association for Research in Personality conference (ARP).
- [WASSA ACL 23 ] **Subbiah\*, M.**, Bhattacharjee\*, A., Hua, Y., Kumarage, T., Liu, H., & McKeown, K. (2023). *Detecting Harmful Agendas in News Articles*. WASSA Workshop at ACL.
- [IWSS ICSWM 21 ] **Subbiah, M.**, & McKeown, K. (2021). *Understanding Identity Signalling in Persuasive Online Text*. International Workshop on Social Sensing at ICWSM.
- [WiML NeurIPS 18 ] Maher, M., **Subbiah, M.**, & Apostoloff, N. (2018). *Cascaded Dataset QA*. Women in Machine Learning at NeurIPS.
- [WiML NeurIPS 18 ] **Subbiah, M.**, lesser, J., & Apostoloff, N. (2018). *Augmenting Training Data with Simulated Images*. Women in Machine Learning at NeurIPS.

## Other Articles and Preprints

- [in submission 25 ] Gupta, M., Varimalla, N. R., Deas, N., **Subbiah, M.**, & McKeown, K. (2025). *AdvSumm: Adversarial Training for Bias Mitigation in Text Summarization*. arXiv.
- [in submission 25 ] Hall, Z., **Subbiah\*, M.**, Zollo\*, T., McKeown, K., & Zemel, R. (2025). *Guiding LLM Decision-Making with Fairness Reward Models*.

## Patents

- [patent 22 ] **Subbiah, M.**, Lesser, J., & Apostoloff, N. (2022). *Training with Simulated Images*. No. US 11,256,958 B1, Apple, Inc.

\*indicates equal contribution

## Teaching

- Spring 2024 • **Guest Lecture, Global Teaching Labs - Uruguay**  
Taught by Yi-Tong Tseo, MIT & University of Montevideo

## Teaching (continued)

- Fall 2023
  - **Teaching Assistant & Guest Lecture**, *Natural Language Generation and Summarization*  
Taught by Kathleen McKeown, Columbia University  
TA Quality Overall: 4.7/5.0
- Summer 2023
  - **Instructor**, *Discrete Mathematics*  
Columbia University  
Course Quality Overall: 4.2/5.0, Instructor Quality Overall: 4.5/5.0
- Spring 2023
  - **Guest Lecture**, *Computational Journalism*  
Taught by Mark Hansen, Columbia University
- Fall 2022
  - **Teaching Assistant & Guest Lecture**, *Natural Language Generation and Summarization*  
Taught by Kathleen McKeown, Columbia University  
TA Quality Overall: 4.9/5.0
- Fall, Spring 2021
  - **Tutor**, *Introductory Computer Science & Discrete Mathematics*  
Columbia University Athletics
- Fall 2014 – Spring 2017
  - **Teaching Assistant**, *Introductory Computer Science and/or Data Structures*  
Williams College

## Invited Speaking

### Research Talks and Posters

- 2024
  - **Talk on "StorySumm: Evaluating Faithfulness in Story Summarization"**  
Summarization track at EMNLP
  - **Poster on "Reading Subtext: Evaluating LLMs on Short Story Summarization with Writers"**  
Summarization track at EMNLP
  - **Talk on "How did we get here?: The rise of large language models and the problem of evaluation"**  
Computer Science Colloquium at Williams College
- 2023
  - **Talk on "Detecting Harmful Agendas in News Articles"**  
WASSA Workshop at ACL
- 2021
  - **Talk on "Understanding Identity Signalling in Persuasive Online Text"**  
International Workshop on Social Sensing at ICWSM
- 2020
  - **Talks on "Language Models are Few-Shot Learners"**  
New York University, seminar series  
Stanford University, NLP seminar series  
Columbia University, NLP seminar series  
Philosophy and Machine Learning Conference at New York University  
G-Research, lecture series
  - **Computer Science colloquium on Language Models**  
Williams College
- 2018
  - **Poster on "Augmenting Training Data with Simulated Images"**  
Women in Machine Learning Workshop at NeurIPS

### Outreach Talks

- 2024
  - **Research Overview: Language Models**  
Stuyvesant High School's research class

## Invited Speaking (continued)

---

- 2023
  - **Half-day interactive seminar on ChatGPT for educators**  
Academy for Teachers' Master Class for 25 high school teachers and librarians
  - **Interview on "Artificial: Episode 2, Selling Out"**  
The Wall Street Journal's "The Journal" podcast (a top 10 daily news podcast in the US)
  - **Talk on "NLP with ChatGPT"**  
Pearson Publishing's AI Catalyst Conference
  - **Talk on "ChatGPT: What is it, how does it work, and what's next?"**  
Columbia University Council of Deans  
Columbia University Faculty
- 2022
  - **Guest for "GPT-3 for Natural Language Processing."**  
SuperDataScience podcast (50,000 listens)
- 2021
  - **Interview on "5 Levels of Difficulty: Machine Learning"**  
Wired Magazine YouTube video (2 million views)
- 2017
  - **Phi Beta Kappa Student Speaker**  
Williams College Commencement (~2,000 attendees)

## Consulting

---

- 2024
  - **Advisor, Early-stage startups**  
Scoping and testing technical requirements for customers of a cost effective green data center solution (BuildAI). Advising on models for educational children's story generation (Nookly).
- 2023
  - **Red Teamer, GPT-4**  
Member of the expert Red Team who helped test GPT-4 for risks pre-release.
  - **Reviewer, Quick Start Guide to Large Language Models**  
Provided expert technical feedback on this guide for professionals written by Sinan Ozdemir and published by Pearson.

## Service

---

- 2025
  - **Reviewer, ACL/COLM/ARR**
- 2024
  - **Reviewer, ACL/EMNLP/COLING/ARR**
- 2023
  - **Panelist, Columbia Engineering School Graduate Council info panel on PhD programs**
- 2022
  - **Reviewer, EMNLP**
  - **Reviewer, In2Writing Workshop @ ACL**
  - **Reviewer, Columbia Pre-submission Application Review program for underrepresented candidates**
- 2021
  - **Co-organizer, Workshop on Enormous Language Models at ICLR**
- 2020
  - **Reviewer, Columbia Pre-submission Application Review program for underrepresented candidates**
  - **Talk, Columbia's "Demystifying the Dissertation" talk series for undergrads**
- 2019
  - **Reviewer, Bay Area Machine Learning symposium**

## Mentoring

---

- 2021 – present
  - **Research Mentor**, *Columbia University research students*
    - Haaris Mian (Masters)
    - Marvin Limpijankit (Masters) → paper in submission, Columbia CS PhD
    - Zara Hall (Undergrad) → paper in submission, Columbia CS Masters
    - Mukur Gupta and Nikhil Varimalla (Masters) → paper in submission, Granica
    - Jacklyn Tsai (Undergrad) → Senior Thesis, Amazon AWS SWE
    - Akankshya Mishra (Masters) → EMNLP paper, paper in submission, Google Gemini MLE
    - Sean Zhang (Masters) → TACL paper (Best Paper Award), CapitalOne SWE
    - Kate Harwood (Masters) → ACL Findings paper, NLP freelancing
    - Adam Storek (Undergrad) → ACL paper, Columbia CS PhD
    - Bobby Yilun Hua (Undergrad) → WASSA workshop paper, Cornell CS PhD
    - Yu-Chen Huang (Undergrad) → Amazon SWE
- 2022
  - **Mentor**, *Williams College CS undergraduate buddy program*
- 2021
  - **Research Mentor**, *Lumiere Education's high school student research program*
- 2020
  - **Program Organizer & Research Mentor**, *OpenAI Scholars (ML transition program)*
- 2019
  - **Mentor**, *Institute of International Education TechWomen program*
- 2016 – 2017
  - **Leader**, *Williams College Underrepresented Identities in CS group*